

Bioactuated Textiles: Exploring Reversible Actuation of Casein Bioplastic in Woven Textiles

Eldy S. Lazaro Vasquez*

ATLAS Institute

University of Colorado Boulder
Boulder, CO, USA

eldy.lazaro@colorado.edu

Viola Arduini*

Critical Media Practices

University of Colorado Boulder
Boulder, CO, USA

viola.arduini@colorado.edu

Etta W Sandry

ATLAS Institute

University of Colorado Boulder
Boulder, CO, USA

etta.sandry@colorado.edu

Srujana Golla

ATLAS Institute

University of Colorado Boulder
Boulder, CO, USA

srujana.golla@colorado.edu

Laura Devendorf

ATLAS Institute & Info Science

University of Colorado Boulder
Boulder, CO, USA

laura.devendorf@colorado.edu

Mirela Alistar

ATLAS Institute & Computer Science

University of Colorado Boulder
Boulder, CO, USA

mirela.alistar@colorado.edu

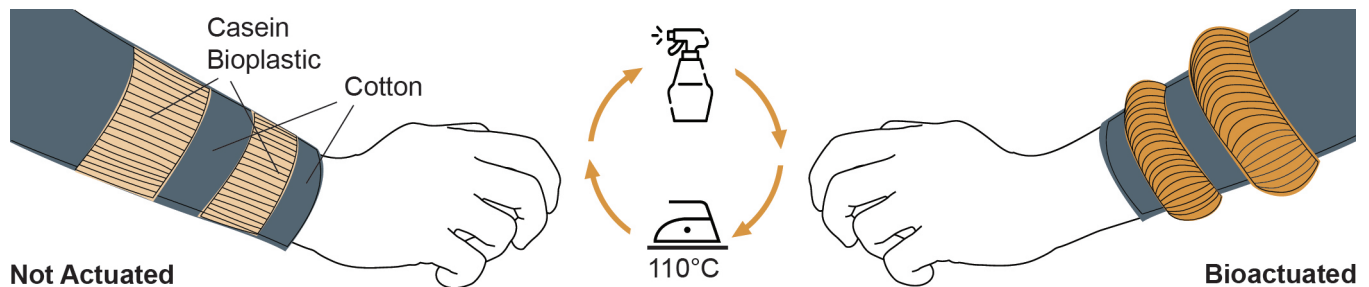


Figure 1: Bioactuation of casein bioplastic-cotton textile. Water mist triggers shape-change (right), ironing reverses it (left).

Abstract

Biomaterials in wearable technology are often explored for their sustainability attributes; this paper broadens that focus to include their functional affordances. We present a casein bioplastic formulation and fabrication technique that preserve its inherent water-triggered shape change. Strips of this biomaterial were integrated into woven swatches to explore how strip width and weave structure influence actuation and textile deformation. We found that the bioactuated textile could be thermally reset, revealing a reversible interaction. We contextualize our findings through envisioned scenarios and discuss how this research can expand interaction modalities in wearable design and interactive textiles.

CCS Concepts

• **Human-centered computing** → **Interaction design process and methods**; *Systems and tools for interaction design*.

*Both authors contributed equally to this research.



This work is licensed under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License.

UbiComp Companion '25, Espoo, Finland

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1477-1/2025/10

<https://doi.org/10.1145/3714394.3750704>

Keywords

bioactuation, biomaterials, weaving, interactive textile, shape-change

ACM Reference Format:

Eldy S. Lazaro Vasquez, Viola Arduini, Etta W Sandry, Srujana Golla, Laura Devendorf, and Mirela Alistar. 2025. Bioactuated Textiles: Exploring Reversible Actuation of Casein Bioplastic in Woven Textiles. In *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*, October 12–16, 2025, Espoo, Finland. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3714394.3750704>

1 Introduction

In wearable technology, interaction emerges through the interplay of materials, electronic components, and how wearables are shaped, layered, or assembled to support functionality. Materials, in particular, enable interactions through soft sensors, shape-changing composites, or thermally responsive threads [9, 11]. While traditionally, such responsiveness has relied on electronic or engineered actuation, a growing body of work in Human-Computer Interaction (HCI) explores how materials themselves, by their composition, can act as responsive elements [25, 40, 43]. Within this shift, biomaterials (living or non-living) have gained attention for their sustainable and interactive qualities [3, 10, 23]. Our work complements these efforts by shifting focus from degradation as a space for interaction [22, 24, 27, 34, 35] to biomaterials' functional affordances.

We explore how the moisture responsiveness of casein bioplastic can be leveraged for interaction when integrated into woven textile structures. We term this shape change in the textile *bioactuation*, as

it is triggered by water or other moisture-rich biological fluids (e.g., sweat) and can be thermally reset by the user for repeated, reversible transformation (see Fig. 1). Rather than designing solutions for specific applications, our motivation is to **broaden interaction possibilities** in wearable and interactive textile design. We do this by leveraging biomaterials' functional affordances, such as shape change, to position them as responsive interfaces that complement existing interaction modalities in the HCI and ISWC communities.

We contribute (1) a simplified casein bioplastic formulation and low-tech fabrication process; (2) an exploration of how bioplastic strip width and weave structure influence textile bioactuation; and (3) discussion and envisioned scenarios of the design possibilities this work opens for wearable and interactive textiles.

2 Materials and Interaction in Wearable Design

As Wiberg [42] argues, materials enable interaction by shaping how users engage with artifacts through their inherent properties and affordances. In wearable design, responsive behaviors such as deformation, color change, or movement are often achieved via electronics or digital control. Increasingly, however, materials themselves, through intrinsic chemical or physical properties, are being explored as actuators. This has led to new interface categories, including those using bio-based or living materials [29]. Building on this shift toward material-centered interactivity, we explore how the inherent moisture responsiveness of casein bioplastic can be modulated through its integration with textile structures. Wearable systems offer a compelling context for this exploration, as their proximity to the body and environment allows material transformations—such as moisture-triggered shape change—to be experienced in situated and embodied ways.

Material Responsiveness in Wearable Systems: Materials with intrinsic responsive behaviors have been integrated into wearables to enable passive actuation or sensing without electronic control. Examples include thermochromic materials changing color with temperature [11], shape-memory alloys altering form under heat [9, 14], and hydrogels swelling in response to moisture [31, 38] or functioning as sensors [38]. These materials support low-complexity designs by leveraging inherent physical and chemical transformations. In wearable systems, textile construction further shapes responsiveness through variations in weave, layering, and fabric composition affecting how actuation manifests at the garment scale [1, 7, 20, 33, 39]. In our work, we followed a material-led design [21] to attune to our material's inherent responsive properties, prioritizing physical behavior as interaction and guiding iterative design across formulation, fabrication, and textile integration.

Interactive Biomaterials in Wearable Interfaces: Building on this material-led approach, we situate our work within the broader space of Sustainable HCI (SHCI), which encompasses diverse approaches to sustainability in interactive systems [5, 6, 32]. A growing body of SHCI research explores biomaterials integration in wearable interfaces. For instance, recent work in Wearable Bio-HCI [45] explores how biomaterials [17, 19, 37], bodily fluids [8, 36, 38, 44, 47], and living matter [4, 26, 46] can enable responsive behaviors in textiles and wearables. In this context, our work complements this line of inquiry by investigating the shape-changing behavior of casein-bioplastic, which enables it to function as a self-actuating element in woven textiles without electronics. Textile structure

further modulates this behavior, expanding design opportunities for tunable bioactuation.

3 Bioactuated Textiles Implementation

This work explores reversible shape-change as a design affordance for wearable textiles by using a casein bioplastic that passively actuates in response to water. We examined how the width of integrated bioplastic strips and variations in weave structure influence the bioactuation of a woven fabric (Fig. 2). This section details the design methodology, biomaterial formulation and fabrication, methods for textile integration, and a preliminary technical evaluation.

3.1 Motivation and Design Approach

This work is motivated by an interest in how biomaterials' moisture or heat responsiveness can support new forms of interaction, temporality, or embodied responsiveness in interactive textiles. Our work is grounded in Research through Design [12], and our material exploration followed a material-led design approach [21]. We attuned to one inherent property of casein bioplastic: its water-triggered curling when wet and stiffening when dried. These observations shaped our design decisions for leveraging this quality. Weaving offered a design space for modulating or amplifying that response, providing spatial control, structural flexibility, and tunability.

3.2 Bioplastic Formulation and Fabrication

Casein is a milk-derived protein found in expired consumer milk and dairy byproducts. While it is possible to precipitate and purify casein from milk directly, for this study, we used commercially available casein to minimize waste during material exploration. We created a simplified casein bioplastic formulation using three ingredients commonly available at grocery stores (Table 1), and developed a low-tech fabrication process that is distinct from historical casein plastics [13], which relied on complex chemical treatments. We tested over 20 ratio variations of this recipe and settled on a 1:10:1 proportion of casein, washing soda solution, and glycerin, which offered a balanced combination of strength, malleability, and flexibility for weaving with the resulting casein bioplastic strips.

Low-tech Fabrication: We start by (1) preparing the washing

Table 1: Casein Bioplastic Formulation

Ingredient	Quantity	Function
Casein (dairy-derived)	25 g	Binder
Washing soda solution	250 mL	Solvent
Glycerin	25 mL	Plasticizer

soda solution by dissolving 2.5 g of it in 250 mL of tap water with low mineral and calcium content. (2) Then, we gradually add 25g of casein powder onto the solution's surface to ensure even suspension without clumping. Using a hand whisk, we gently incorporate the casein powder in the solution to minimize air entrapment. We stir until the mixture reaches a semi-solid, viscous consistency. (3) Next, we add 25 mL of glycerin and heat the mixture in an 85°C water bath for 15 minutes, stirring every 3 minutes. We skim off the top foam with a spatula, and sieve the solution. After a second 5-minute heating, the mixture is cooled in an ice bath until it reaches approx. 45–55°C (2–2.5 minutes). A surface film is removed, and the solution is sieved again. (4) We pour the mixture into a flat 24 cm × 29 cm silicone mold to cure at room temperature (22–25 °C) for 5–6

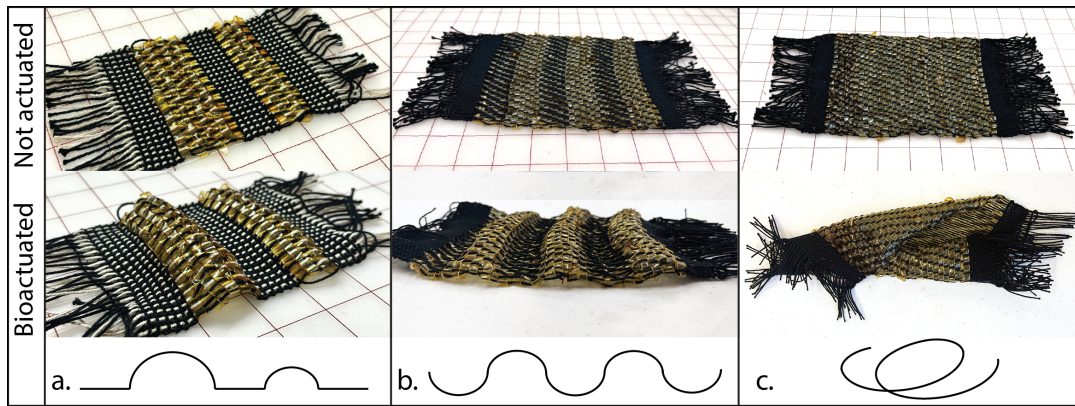


Figure 2: Textile samples before (top) and after (bottom) bioactuation, with cotton warp and 5 mm bioplastic strips weft integrated using 1/3 twill structure: (a) two stripe widths of 25 mm and 38 mm separated by tabby; (b) 25 mm stripes with alternating twill face (front/back); (c) full length integration.

days in a low-humidity environment, resulting in a 0.5 mm thick translucent bioplastic sheet with a yellow tint, ready to laser cut.

Bioplastic Mass Loss During Drying Process: Drying is a critical step in bioplastic prototyping and influences mechanical properties, dimensional stability, and integration readiness (e.g., cutting, weaving, or adhesion). We measured the mass reduction of bioplastic sheets during the drying process (88% avg), observing a proportional relationship between mass loss, drying time, and initial material quantity. This behavior, resulting from water evaporation, offered a practical metric for developing a consistent drying protocol tailored to different batch sizes, helping us to standardize material preparation before textile integration. We found that 0.5 to 1 mm bioplastic thickness was stable, malleable, and easier to laser cut and manipulate during textile integration.

3.3 Textile Integration for Bioactuation

When exposed to water, the casein bioplastic did not dissolve as other biomaterials do [2, 23, 24]; instead, it curled and became firmer and more rigid (Fig. 3). Narrow, laser-cut strips maintained their cohesion and responded visibly to moisture. To better understand this responsiveness, we tested different forms of water exposure, including misting, dripping, and full immersion. Misting produced the most pronounced and consistent actuation: the strips curled and stiffened within seconds and dried quickly. In contrast, dripping and immersion saturated the material, which limited the extent and visibility of the shape change. Based on these results, we focused on integrating the bioplastic into woven textiles using misting as the activation method. To explore how textile structure influences this response, we wove a series of swatches using a 4-harness floor loom and an 8-harness tabletop loom, both with 8/4 cotton in the warp. All swatches followed the same weave pattern, similar to Figure 2a, with two bioplastic stripes (13 mm and 25 mm wide) where different

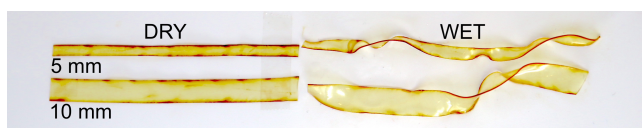


Figure 3: Shape-change of casein bioplastic strips.

weave structures were tested using three bioplastic strip widths: 3 mm and 5 mm. We used tabby to separate these testing areas. For 3 mm bioplastic strips, we tested 1/3 twill, 1/3 twill back, 2/6 twill, basket, tabby, and satin. For 5 mm strips, we tested 1/3 twill, tabby, and one sample with tabby alternating cotton and bioplastic. We wove a separate set of swatches, made with 5 mm strips of bioplastic, in various patterns (8/4 satin, 9/3 satin, 7/1 satin, 11/1 satin, and waffle) on a TC2 digital jacquard loom with a finer warp, allowing comparison across weaving setups. Our full documentation and results can be found at <https://shorturl.at/aG1ph>. These exploratory tests helped identify a combination of 1/3 twill structure with 5 mm bioplastic strips as the most consistent and visibly responsive combination. Based on this, we designed a second set of swatches to focus on variations within twill, described in the next section.

3.4 Preliminary Technical Evaluation

Building on the results of our initial integration tests, we conducted a focused evaluation of twill-based structures using 5 mm and 10 mm bioplastic strips. This second phase explored how weave structure and strip width influence bioactuation behavior. All swatches measured approximately 10 × 15 cm and used 8/4 cotton warp with laser-cut bioplastic weft.

Twill-Based Variation Testing. Figure 2 shows three configurations, all woven with 5 mm bioplastic strips: In Fig. 2a, we wove two bioplastic stripes (25 mm and 38 mm wide), both in 1/3 twill separated by tabby, and tested how stripe width affects the degree of curvature. In Fig. 2b, we wove two adjacent bioplastic stripes in alternating twill directions (1/3 and 3/1) to explore whether reversing the twill face produces bioactuation on both fabric sides. In Fig. 2c, we wove continuous bioplastic across the full swatch in 1/3 twill to evaluate full-surface bioactuation. In addition to these swatches, we tested 10 mm strips using the same weave structure as Fig. 2a and Fig. 2b to test the limits of curvature and structural stability at larger widths of bioplastic.

Bioactuation Response. Across all configurations, 5 mm strips offered the most reliable balance between bioactuation strength and weave stability. While the 10 mm strips responded strongly to moisture, their greater width made them less structurally stable in some weaves, and difficult to weave with. For example, in the

same weave configuration as Fig. 2a, the wider stripes had difficulty holding structure. In the configuration shown in Fig. 2b, the 10 mm strips performed more reliably, though the resulting fabric was noticeably coarser to the touch. Among the three configurations, the full twill coverage in Fig. 2c actuated most quickly, curling within a minute. The partial twill configurations (Fig. 2a and b) actuated more gradually, within 4 and 9 minutes, respectively. No visible actuation was observed in samples woven with finer warp yarns on the TC2 loom, likely due to the warp's higher density constraining movement.

Reversibility. We tested tension, heat, and weight as methods to reverse the textile bioactuation. Tension alone didn't visibly flatten the textile, but heat and weight did, with heat yielding faster, more consistent results. We found the bioplastic to be thermo-reversible, with 110°C for 30 seconds being most effective. We used a household iron on its lowest setting, with a cotton cloth placed between the iron and the textile. This process restored the material to its pre-moistured state. We repeated this test for five cycles on the same swatches (Fig. 2), finding no visible deterioration of the bioplastic.

4 Discussion through Envisioned Scenarios

This research explores how bioactuation—moisture-triggered shape change in textile systems—can be enabled through the integration of casein bioplastic. By varying bioplastic strip width and weave structure, we observed how this transformation could be tuned across the textile surface. These findings position textile construction itself as a site of interaction design: a programmable, physical system for shaping how materials respond and behave in context.

This approach extends prior work on sensing and actuation in textiles [18, 30, 41] by demonstrating how structure and material responsiveness can be composed to shape interaction through fabrication itself. Rather than treating responsiveness as something to be added through electronics or sensing layers, we show how it can emerge from the composition of textile and biomaterial properties. While our tests were at the swatch scale, they suggest how familiar textile techniques can offer precise, physical control over shape change in wearable systems—opening up interaction modalities that are low-tech, reversible, and materially grounded.

Sleeve Accent as Embodied Feedback: In this scenario, strips of casein bioplastic are integrated into a decorative sleeve accent. When the sleeve is exposed to moisture, through sweat or environmental humidity, the strips bend outward, revealing a hidden color or material beneath (Fig. 1). This interaction is slow and situated. The visible transformation reflects contextual change in the body or environment, offering a form of embodied feedback. The actuation is expressive, but grounded in material behavior—where the garment participates in environmental sensing through its form. This aligns with work in HCI that foregrounds material responsiveness as a legitimate and expressive form of interactivity [16]—expanding toward body-situated behaviors shaped by the affordances of biomaterials themselves [15].

Material Memory in Collar and Waistband: This scenario embeds bioplastic strips into the collar and waistband of a woven garment. In a moisture-rich setting such as a greenhouse or rainy outdoor environment, the bioactuated regions respond by curling or lifting: the collar opens slightly for ventilation, while the waistband develops a rippling effect. These changes remain after a few hours,

leaving a visible trace of prior conditions. To reset, the wearer applies heat using a household iron, restoring the textile to its original state. This reset is not automatic but intentional, introducing a small moment of care, attention, or aesthetic decision. Reversibility becomes part of the interaction, shaped by material temporality and user engagement rather than just a functional reset.

These envisioned scenarios show how interaction can emerge from the material behavior of the textile system itself. Bioactuation unfolds slowly and physically, shaped by the weave structure, the placement of bioplastic, and the conditions of wear. It invites interaction that is reversible and situated—where changes are felt and seen, rather than read or sensed through digital signals. Resetting becomes part of this engagement: not automatic, but a deliberate act performed by the wearer using everyday tools like a household iron. This low-tech reversibility supports repair, adaptation, and ongoing participation, aligning with sustainable interaction design practices that emphasize care, material longevity, and user agency.

While this work focused on woven swatches with bioplastic in strip form, future explorations could involve yarn- or fiber-level fabrication enabled by existing open-source tools [22, 28, 48]. This may allow more continuous integration across larger garments and enable deformation to emerge not just in isolated regions but as a distributed property of the textile. We also recognize that scaling up introduces challenges around consistency, wearability, and long-term durability. In these contexts, hybrid systems—combining material responsiveness with computational control—may support more layered forms of interaction that blend environmental reactivity with contextual adaptation. Finally, this work expands ongoing conversations in Wearable Bio-HCI [45] by framing biomaterials not only as sustainable substrates but as active components in interactive systems. Moisture is not interpreted by a sensor but enacted through the textile's shape. This shift—from sensing to shaping—positions bioactuation as a materially grounded, temporally rich mode of interaction that unfolds through the rhythms of wear, environment, and care.

5 Conclusion and Future Work

This work introduces a casein-based bioplastic and a low-tech fabrication process that preserves its reversible, moisture-triggered shape change. Through textile integration, we show how shape change can be physically tuned by adjusting parameters such as strip width and weave structure. Rather than embedding interactivity through computation, this approach demonstrates how material responsiveness and textile construction together can support alternative interaction modalities—functional, expressive, and low in complexity—for wearable systems.

Future work will extend this approach to full garments, where questions of consistency, wearability, and long-term performance come into focus. We are also interested in exploring additional environmental triggers such as temperature or pH. We would also like to explore more symbolic and embodied dimensions of wearable systems, including themes of care, leakage, or bodily fluids.

Acknowledgments

This work was supported by the National Science Foundation under Grant #2413631. We thank the BTU Lab for equipment access.

References

- [1] Hossein Baniasadi, Zahra Madani, Mithila Mohan, Maija Vaara, Sami Lipponen, Jaana Vapaavuori, and Jukka V. Seppälä. 2023. Heat-induced actuator fibers: Starch-containing biopolyamide composites for functional textiles. *ACS Applied Materials & Interfaces* 15, 41 (2023), 48584–48600.
- [2] Fiona Bell, Latifa Al Naimi, Ella McQuaid, and Mirela Alistar. 2022. Designing with Alganyl. In *Proceedings of the Sixteenth International Conference on Tangible, Embedded, and Embodied Interaction* (Daejeon, Republic of Korea) (TEI '22). Association for Computing Machinery, New York, NY, USA, Article 2, 14 pages. doi:10.1145/3490149.3501308
- [3] Fiona Bell, Erin McClure, Camila Friedman-Gerlicz, Ruby Ta, and Leah Buechley. 2024. Shape-Changing Clay-Dough: Taking a Material-Oriented Approach to 3D Printing Ceramic Forms. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 864, 19 pages. doi:10.1145/3613904.3642246
- [4] Fiona Bell, Michelle Ramsahoye, Joshua Coffie, Julia Tung, and Mirela Alistar. 2023. µMe: Exploring the Human Microbiome as an Intimate Material for Living Interfaces. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference* (Pittsburgh, PA, USA) (DIS '23). Association for Computing Machinery, New York, NY, USA, 2019–2033. doi:10.1145/3563657.3596133
- [5] Nicola Besana, Peter Gall Krogh, Marianne Graves Petersen, and Davide Spallazzo. 2024. Expanding the Concept of Sustainable Interaction Design: A Systematic Review. *Sustainability* 16, 17 (Jan. 2024), 7486. doi:10.3390/su16177486 Number: 17 Publisher: Multidisciplinary Digital Publishing Institute.
- [6] Christina Bremer, Bran Knowles, and Adrian Friday. 2022. Have We Taken On Too Much?: A Critical Review of the Sustainable HCI Landscape. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 41, 11 pages. doi:10.1145/3491102.3517609
- [7] Alice Buso, Holly Mcquillan, Kaspar Jansen, and Elvin Karana. 2024. AnimaTo: Designing a Multimorphic Textile Artefact for Performativity. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference* (Copenhagen, Denmark) (DIS '24). Association for Computing Machinery, New York, NY, USA, 20–34. doi:10.1145/3643834.3660738
- [8] Nadia Campo Woytuk, Joo Young Park, Jan Maslik, Marianela Cioffi Felice, and Madeline Balaam. 2023. Tactful Feminist Sensing: Designing for Touching Vaginal Fluids. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference* (Pittsburgh, PA, USA) (DIS '23). Association for Computing Machinery, New York, NY, USA, 2642–2656. doi:10.1145/3563657.3595966
- [9] George Chernyshov, Benjamin Tag, Cedric Caremel, Feier Cao, Gemma Liu, and Kai Kunze. 2018. Shape memory alloy wire actuators for soft, wearable haptic devices. In *Proceedings of the 2018 ACM International Symposium on Wearable Computers* (Singapore, Singapore) (ISWC '18). Association for Computing Machinery, New York, NY, USA, 112–119. doi:10.1145/3267242.3267257
- [10] Karijn Den Teuling, Amy Winters, and Miguel Bruns. 2024. Chitosan Biofilm Actuators: Humidity Responsive Materials for Sustainable Interaction Design. In *Proceedings of the Eighteenth International Conference on Tangible, Embedded, and Embodied Interaction* (Cork, Ireland) (TEI '24). Association for Computing Machinery, New York, NY, USA, Article 81, 7 pages. doi:10.1145/3623509.3635262
- [11] Laura Devendorf and Chad Di Lauro. 2019. Adapting Double Weaving and Yarn Plying Techniques for Smart Textiles Applications. In *Proceedings of the Thirteenth International Conference on Tangible, Embedded, and Embodied Interaction* (Tempe, Arizona, USA) (TEI '19). Association for Computing Machinery, New York, NY, USA, 77–85. doi:10.1145/3294109.3295625
- [12] William Gaver. 2012. What should we expect from research through design?. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Austin, Texas, USA) (CHI '12). Association for Computing Machinery, New York, NY, USA, 937–946. doi:10.1145/2207676.2208538
- [13] Marianne Gilbert. 2017. Chapter 1 - plastics materials: Introduction and historical development. In *Brydson's Plastics Materials (Eighth Edition)*, Marianne Gilbert (Ed.). Butterworth-Heinemann, 1–18. doi:10.1016/B978-0-323-35824-8.00001-3
- [14] Alireza Golgouneh, Brad Holschuh, and Lucy Dunne. 2020. A Controllable Biomimetic SMA-actuated Robotic Arm. In *2020 8th IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechanics (BioRob)*, 152–157. doi:10.1109/BioRob49111.2020.9224371
- [15] Phillip Gough, Soojeong Yoo, Martin Tomitsch, and Naseem Ahmadpour. 2021. Applying Bioaffordances through an Inquiry-Based Model: a literature review of interactive biodesign. *International Journal of Human-Computer Interaction* 37, 17 (2021), 1583–1597.
- [16] Shad Gross, Jeffrey Bardzell, and Shaowen Bardzell. 2014. Structures, forms, and stuff: the materiality and medium of interaction. *Personal and Ubiquitous Computing* 18, 3 (2014), 637–649.
- [17] Sofia Guridi. 2025. Crafting Sustainable Interfaces: The Aesthetic Potential of Bio-based eTextiles. In *Companion Publication of the 2025 ACM Designing Interactive Systems Conference* (DIS '25 Companion). Association for Computing Machinery, New York, NY, USA, 446–451. doi:10.1145/3715668.3736340
- [18] Sofia Guridi, Fevzihan Basarir, Mijung Cho, An Nguyen, Jaana Vapaavuori, and Michael Hummel. 2024. Sense and Degrade: Fully Bio-Based and Biodegradable eTextile Capacitive Sensor. In *2024 International Conference on the Challenges, Opportunities, Innovations and Applications in Electronic Textiles (E-Textiles)*. IEEE, 118–125.
- [19] Sofia Guridi, Matteo Iannacchero, and Emmi Pouta. 2024. Towards More Sustainable Interactive Textiles: A Literature Review on The Use of Biomaterials for eTextiles. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–19.
- [20] Min-Woo Han, Min-Soo Kim, and Sung-Hoon Ahn. 2020. Shape memory textile composites with multi-mode actuations for soft morphing skins. *Composites Part B: Engineering* 198 (2020), 108170.
- [21] Elvin Karana, Bahareh Barati, Valentina Rognoli, Anouk Zeeuw Van Der Laan, et al. 2015. Material driven design (MDD): A method to design for material experiences. *International journal of design 9*, 2 (2015), 35–54.
- [22] Eldy S. Lazaro Vasquez, Mirela Alistar, Laura Devendorf, and Michael L. Rivera. 2024. Desktop Biofibers Spinning: An Open-Source Machine for Exploring Biobased Fibers and Their Application Towards Sustainable Smart Textile Design. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 856, 18 pages. doi:10.1145/3613904.3642387
- [23] Eldy S. Lazaro Vasquez, Lily M Gabriel, Mikhaila Friske, Shanel Wu, Sasha De Koninck, Laura Devendorf, and Mirela Alistar. 2023. Designing Dissolving Wearables. In *Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing* (Cancun, Quintana Roo, Mexico) (UbiComp/ISWC '23 Adjunct). Association for Computing Machinery, New York, NY, USA, 286–290. doi:10.1145/3594739.3610781
- [24] Eldy S. Lazaro Vasquez, Netta Ofer, Shanel Wu, Mary Etta West, Mirela Alistar, and Laura Devendorf. 2022. Exploring Biofoam as a Material for Tangible Interaction. In *Proceedings of the 2022 ACM Designing Interactive Systems Conference* (Virtual Event, Australia) (DIS '22). Association for Computing Machinery, New York, NY, USA, 1525–1539. doi:10.1145/3532106.3533494
- [25] Xin Liu, Chengkuo Lee, Clement Zheng, and Ching Chiuan Yen. 2025. Designing Physical Interactions with Triboelectric Material Sensing. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI '25). Association for Computing Machinery, New York, NY, USA, Article 683, 20 pages. doi:10.1145/3706598.3714194
- [26] Jasmine Lu and Pedro Lopes. 2022. Integrating Living Organisms in Devices to Implement Care-based Interactions. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (Bend, OR, USA) (UIST '22). Association for Computing Machinery, New York, NY, USA, Article 28, 13 pages. doi:10.1145/3526113.3545629
- [27] Qiuyu Lu, Semina Yi, Mengtian Gan, Jihong Huang, Xiao Zhang, Yue Yang, Chenyi Shen, and Lining Yao. 2024. Degrade to Function: Towards Eco-friendly Morphing Devices that Function Through Programmed Sequential Degradation. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology* (Pittsburgh, PA, USA) (UIST '24). Association for Computing Machinery, New York, NY, USA, Article 109, 24 pages. doi:10.1145/3654777.3676464
- [28] Malu Lücking, MOME Innovation, Ferenc Kovacs-Nagy, and Nora Gulya. 2025. The Wet Biofibre Spinner. <https://www.instagram.com/designstudio.malu/>
- [29] Timothy Merritt, Foad Hamidi, Mirela Alistar, and Marta DeMenezes. 2020. Living media interfaces: a multi-perspective analysis of biological materials for interaction. *Digital Creativity* 31, 1 (2020), 1–21.
- [30] Emmi Pouta, Jussi Ville Mikkonen, and Antti Salovaara. 2024. Opportunities with Multi-Layer Weave Structures in Woven E-Textile Design. *ACM Trans. Comput.-Hum. Interact.* 31, 5, Article 62 (Nov. 2024), 38 pages. doi:10.1145/3689039
- [31] Michael L. Rivera, Jack Forman, Scott E. Hudson, and Lining Yao. 2020. Hydrogel-Textile Composites: Actuators for Shape-Changing Interfaces. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI EA '20). Association for Computing Machinery, New York, NY, USA, 1–9. doi:10.1145/3334480.3382788
- [32] Sabrina Scuri, Marta Ferreira, Nuno Jardim Nunes, Valentina Nisi, and Cathy Mulligan. 2022. Hitting the Triple Bottom Line: Widening the HCI Approach to Sustainability. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 332, 19 pages. doi:10.1145/3491102.3517518
- [33] Amanda Skalitzy, Austin Gurley, David Beale, and Kyle Kubik. 2018. Design and analysis of SMA woven fabric. In *Smart Materials, Adaptive Structures and Intelligent Systems*, Vol. 51944. American Society of Mechanical Engineers, V001T03A033.
- [34] Katherine W Song, Aditi Maheshwari, Eric M Gallo, Andreea Danielescu, and Eric Paulos. 2022. Towards decomposable interactive systems: Design of a backyard-degradable wireless heating interface. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–12.
- [35] Katherine W Song and Eric Paulos. 2021. Unmaking: Enabling and celebrating the creative material of failure, destruction, decay, and deformation. In *Proceedings*

- of the 2021 CHI Conference on Human Factors in Computing Systems. 1–12.
- [36] Shuyi Sun, Alejandra Ruiz, Sima Pirmoradi, and Katia Vega. 2023. BioSparks: Jewelry as Electrochemical Sweat Biosensors with Modular, Repurposing and Interchangeable Approaches. In *Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing*. 315–320.
- [37] Eldy S. Lazaro Vasquez, Sepideh Mohammadi, Latifa Al Naimi, Shira David, and Mirela Alistar. 2025. Bio-e-Nails: a Sustainable Design Approach to Biobased Nail Interfaces. In *Proceedings of the Nineteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '25)*. Association for Computing Machinery, New York, NY, USA, Article 46, 15 pages. doi:10.1145/3689050.3704426
- [38] Eldy S. Lazaro Vasquez, Ali K. Yetisen, and Katia Vega. 2020. BraceIO: biosensing through hydrogel dental ligatures. In *Proceedings of the 2020 ACM International Symposium on Wearable Computers (Virtual Event, Mexico) (ISWC '20)*. Association for Computing Machinery, New York, NY, USA, 87–89. doi:10.1145/3410531.3414290
- [39] Yvonne Y. F. Chan Vili. 2007. Investigating Smart Textiles Based on Shape Memory Materials. *Textile Research Journal* 77, 5 (2007), 290–300. doi:10.1177/0040517507078794 arXiv:https://doi.org/10.1177/0040517507078794
- [40] Michael R. Vinciguerra, Dinesh K. Patel, Wuzhou Zu, Mahmoud Tavakoli, Carmel Majidi, and Lining Yao. 2023. Multimaterial Printing of Liquid Crystal Elastomers with Integrated Stretchable Electronics. *ACS Applied Materials & Interfaces* 15, 20 (May 2023), 24777–24787. doi:10.1021/acsami.2c23028 Publisher: American Chemical Society.
- [41] Kathryn Walters, Laura Devendorf, and Karin Landahl. 2024. Animated Linen: Using High-twist Hygromorphic Yarn to Produce Interactive Woven Textiles. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference (Copenhagen, Denmark) (DIS '24)*. Association for Computing Machinery, New York, NY, USA, 1177–1191. doi:10.1145/3643834.3662146
- [42] Mikael Wiberg, Hiroshi Ishii, Paul Dourish, Anna Vallgård, Tobie Kerridge, Petra Sundström, Daniela Rosner, and Mark Rolston. 2013. Materiality matters—experience materials. *Interactions* 20, 2 (March 2013), 54–57. doi:10.1145/2427076.2427087
- [43] Lining Yao, Jifei Ou, Chin-Yi Cheng, Helene Steiner, Wen Wang, Guanyun Wang, and Hiroshi Ishii. 2015. bioLogic: Natto Cells as Nanoactuators for Shape Changing Interfaces. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems (Seoul, Republic of Korea) (CHI '15)*. Association for Computing Machinery, New York, NY, USA, 1–10. doi:10.1145/2702123.2702611
- [44] Meredith Young-Ng, Grace Chen, Danielle Kiyama, Anna-Sofia Giannicola, Erkin Şeker, and Katia Vega. 2023. Sweatcessory: a Wearable Necklace for Sensing Biological Data in Sweat. In *Adjunct Proceedings of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing and the 2022 ACM International Symposium on Wearable Computers (Cambridge, United Kingdom) (UbiComp/ISWC '22 Adjunct)*. Association for Computing Machinery, New York, NY, USA, 141–143. doi:10.1145/3544793.3560342
- [45] Jingwen Zhu, Fiona Bell, Katherine W Song, Katia Vega, Aditya Shekhar Nittala, Mirela Alistar, Leah Buechley, and Cindy Hsin-Liu Kao. 2025. Wearable Bio-HCI: Challenges & Opportunities. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*. Association for Computing Machinery, New York, NY, USA, Article 866, 5 pages. doi:10.1145/3706599.3716295
- [46] Jingwen Zhu, Samantha Chang, Ruth Zhao, and Cindy Hsin-Liu Kao. 2025. LivingLoom: Investigating Human-Plant Symbiosis Through Integrating Living Plants Into (E-)Textiles. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 773, 18 pages. doi:10.1145/3706598.3713156
- [47] Jingwen Zhu, Nadine El Nesr, Christina Simon, Nola Rettenmaier, Kaitlyn Beiler, and Cindy Hsin-Liu Kao. 2023. BioWeave: Weaving Thread-Based Sweat-Sensing On-Skin Interfaces. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (San Francisco, CA, USA) (UIST '23)*. Association for Computing Machinery, New York, NY, USA, Article 35, 11 pages. doi:10.1145/3586183.3606769
- [48] Jingwen Zhu, Megan Wu, Ruth Zhao, Samantha Chang, and Cindy Hsin-Liu Kao. 2025. Empowering Sustainable E-Textiles: DIY Biofiber Wet Spinning for Community Material Exploration. In *Proceedings of the 2025 ACM Designing Interactive Systems Conference (DIS '25)*. Association for Computing Machinery, New York, NY, USA, 2395–2409. doi:10.1145/3715336.3735409